```
iris = load_iris()
X = iris.data
y = iris.target

# Convert to DataFrame (optional but useful)
df = pd.DataFrame(X, columns=iris.feature_names)
df['target'] = y

print(df.head())

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

print("Preprocessing Done ✅ ")
```

🔹 2. Linear Regression (Not ideal but included as per list)

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("MSE:", mean_squared_error(y_test, y_pred))
```

🔹 3. Logistic Regression

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
```

---

4. KNN

```
from sklearn.neighbors import KNeighborsClassifier

model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("KNN Accuracy:", accuracy_score(y_test, y_pred))
```

---

5. SVM

```
from sklearn.svm import SVC

model = SVC(kernel='linear')
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("SVM Accuracy:", accuracy_score(y_test, y_pred))
```

---

6. Naive Bayes

```
from sklearn.naive_bayes import GaussianNB

model = GaussianNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Naive Bayes Accuracy:", accuracy_score(y_test, y_pred))
```

---

7. AdaBoost

```
from sklearn.ensemble import AdaBoostClassifier

model = AdaBoostClassifier(n_estimators=50, random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("AdaBoost Accuracy:", accuracy_score(y_test, y_pred))
```

---

8. K-Means Clustering (Unsupervised)

```
from sklearn.cluster import KMeans

kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(X)

labels = kmeans.labels_

print("Cluster Labels:", labels[:10])
```

🔥 **Bonus: Accuracy Comparison (All models together)**

```
models = {
    "Logistic": LogisticRegression(max_iter=200),
    "KNN": KNeighborsClassifier(),
    "SVM": SVC(),
    "Naive Bayes": GaussianNB(),
    "AdaBoost": AdaBoostClassifier()
}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    print(name, "Accuracy:", accuracy_score(y_test, y_pred))
```